

# Predictive Analytics - Assignment 4

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## PS-3

### Q.5

**Problem to demonstrate the utility of nonlinear regression over linear regression**

**Get the fgl data set from “MASS” library**

```
library(MASS)
data("fgl")
str(fgl)

## 'data.frame':    214 obs. of  10 variables:
## $ RI : num  3.01 -0.39 -1.82 -0.34 -0.58 ...
## $ Na : num  13.6 13.9 13.5 13.2 13.3 ...
## $ Mg : num  4.49 3.6 3.55 3.69 3.62 3.61 3.6 3.61 3.58 3.6 ...
## $ Al : num  1.1 1.36 1.54 1.29 1.24 1.62 1.14 1.05 1.37 1.36 ...
## $ Si : num  71.8 72.7 73 72.6 73.1 ...
## $ K : num  0.06 0.48 0.39 0.57 0.55 0.64 0.58 0.57 0.56 0.57 ...
## $ Ca : num  8.75 7.83 7.78 8.22 8.07 8.07 8.17 8.24 8.3 8.4 ...
## $ Ba : num  0 0 0 0 0 0 0 0 0 0 ...
## $ Fe : num  0 0 0 0 0 0.26 0 0 0 0.11 ...
## $ type: Factor w/ 6 levels "WinF","WinNF",...: 1 1 1 1 1 1 1 1 1 1 ...

head(fgl)

##      RI      Na      Mg      Al      Si      K      Ca      Ba      Fe type
## 1  3.01  13.64  4.49  1.10  71.78  0.06  8.75   0  0.00 WinF
## 2 -0.39  13.89  3.60  1.36  72.73  0.48  7.83   0  0.00 WinF
## 3 -1.82  13.53  3.55  1.54  72.99  0.39  7.78   0  0.00 WinF
## 4 -0.34  13.21  3.69  1.29  72.61  0.57  8.22   0  0.00 WinF
## 5 -0.58  13.27  3.62  1.24  73.08  0.55  8.07   0  0.00 WinF
## 6 -2.04  12.79  3.61  1.62  72.97  0.64  8.07   0  0.26 WinF
```

The dataset contains:

RI → Refractive Index (response)

Na → Sodium

Mg → Magnesium

Al → Aluminium

Si → Silicon

K → Potassium

Ca → Calcium

Ba → Barium

Fe → Iron

type → glass type (we focus on Vehicle Window glass)

```
fgl_veh = subset(fgl, fgl$type == "Veh")
head(fgl_veh)
```

```
##      RI      Na      Mg      Al      Si      K      Ca      Ba      Fe      type
## 147 -0.31 13.65 3.66 1.11 72.77 0.11 8.60 0 0.00 Veh
## 148 -1.90 13.33 3.53 1.34 72.67 0.56 8.33 0 0.00 Veh
## 149 -1.30 13.24 3.57 1.38 72.70 0.56 8.44 0 0.10 Veh
## 150 -1.57 12.16 3.52 1.35 72.89 0.57 8.53 0 0.00 Veh
## 151 -1.35 13.14 3.45 1.76 72.48 0.60 8.38 0 0.17 Veh
## 152  3.27 14.32 3.90 0.83 71.50 0.00 9.49 0 0.00 Veh
```

- (a) Considering the refractive index (RI) of “Vehicle Window glass” as the variable of interest and assuming linearity of regression, run multiple linear regression of RI on different metallic oxides. From the p value, report which metallic oxide best explains the refractive index.

We regress:  $RI = \beta_0 + \beta_1 Na + \beta_2 Mg + \beta_3 Al + \beta_4 Si + \beta_5 K + \beta_6 Ca + \beta_7 Ba + \beta_8 Fe + \epsilon$

```
model_a = lm(RI ~ Na + Mg + Al + Si + K + Ca + Ba + Fe, data = fgl_veh)
summary(model_a)
```

```
##
## Call:
## lm(formula = RI ~ Na + Mg + Al + Si + K + Ca + Ba + Fe, data = fgl_veh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29194 -0.08582  0.00072  0.10740  0.33524
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 131.4641    47.2669   2.781  0.02388 *
## Na          -0.4333     0.3509  -1.235  0.25190
## Mg          -0.2866     1.0075  -0.285  0.78325
## Al          -0.8909     0.5550  -1.605  0.14713
## Si          -1.8824     0.4993  -3.770  0.00547 **
## K           -2.4232     0.9725  -2.492  0.03743 *
## Ca           1.5326     0.5818   2.634  0.02998 *
## Ba           0.3517     2.6904   0.131  0.89922
## Fe           3.8931     0.9581   4.063  0.00362 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2621 on 8 degrees of freedom
## Multiple R-squared:  0.9906, Adjusted R-squared:  0.9813
## F-statistic: 105.9 on 8 and 8 DF, p-value: 2.622e-07
```

## Interpretation

We know that the element with lowest p-value and highest t-statistic value will be the best predictor.

Here we observe that “Fe” (Iron) is the best oxide for predicting the refractive index of vehicle glass window.

(b) Run a simple linear regression of RI on the best predictor chosen in (a).

$$RI = \beta_0 + \beta_1 Fe + \epsilon$$

```
model_b = lm(RI ~ Fe, data = fgl_veh)
summary(model_b)

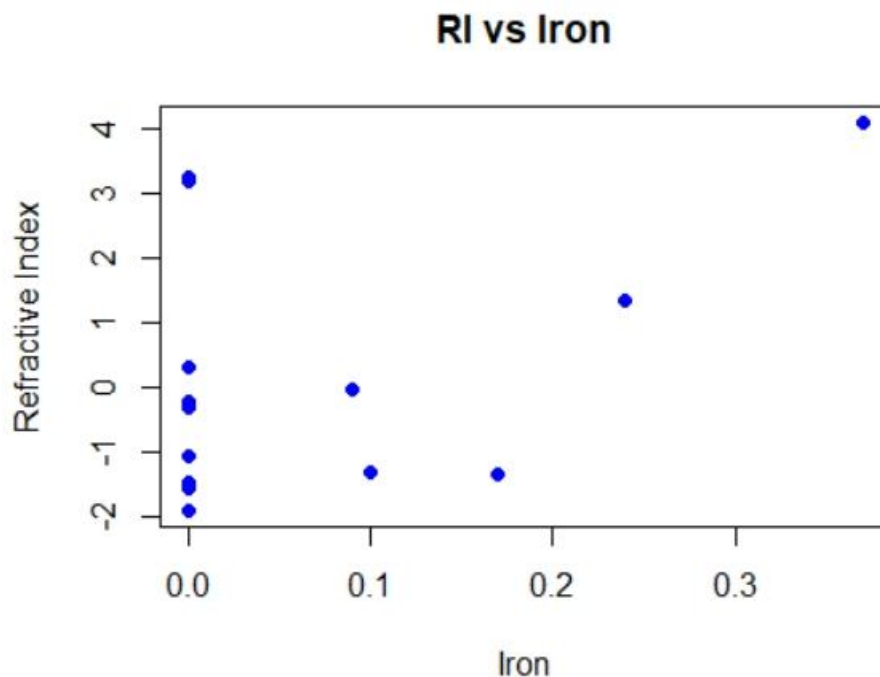
##
## Call:
## lm(formula = RI ~ Fe, data = fgl_veh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.2324 -1.0693 -0.2715  0.2907  3.7707
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5007     0.4861  -1.030   0.3193
## Fe             8.1362     4.0780   1.995   0.0645 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.759 on 15 degrees of freedom
## Multiple R-squared:  0.2097, Adjusted R-squared:  0.157
## F-statistic: 3.981 on 1 and 15 DF, p-value: 0.06452
```

## Interpretation

The  $R^2$  value of this model is 0.2097, hence we conclude that the oxide “Fe” (Iron) explains 20.97% variability in the Refractive Index (RI) of the vehicle glass window.

(c) Can you further improve the regression of the refractive index of “Vehicle Window glass” on the predictor chosen by you in part (a)? Give the new fitted model and compare its performance with the model in (b).

```
# Plotting to check the relationship between RI and Fe
plot(fgl_veh$Fe, fgl_veh$RI,
     pch=19, col="blue",
     main="RI vs Iron",
     xlab="Iron",
     ylab="Refractive Index")
```



### Interpretation

Here, we observe that there is no linear relationship between RI and Fe, rather we see a curvy relation between the two, hence we try fitting a quadratic regression to improve the fit.

$$RI = \beta_0 + \beta_1 Fe + \beta_2 Fe^2 + \epsilon$$

```
Quad_model = lm(RI ~ Fe + I(Fe^2), data= fgl_veh)
summary(Quad_model)
```

```
##
## Call:
## lm(formula = RI ~ Fe + I(Fe^2), data = fgl_veh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6215 -1.1715 -0.1345  0.5985  3.5485
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.2785     0.4712  -0.591   0.564
## Fe           -12.1810    12.0408  -1.012   0.329
## I(Fe^2)       65.9600    37.0798   1.779   0.097 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.645 on 14 degrees of freedom
## Multiple R-squared:  0.3554, Adjusted R-squared:  0.2633
## F-statistic:  3.86 on 2 and 14 DF,  p-value: 0.04623
```

## Interpretation

The  $R^2$  value of this Quadratic model is 0.3554, hence we see that this model explains 35.54% variability on refractive index of vehicle glass window.

```
summary(model_b)$r.squared
## [1] 0.2097192

summary(Quad_model)$r.squared
## [1] 0.355413
```

## Interpretation

Hence we see a 14.57% increase in the explanation of variability of Iron on Refractive Index of Vehicle Glass Window.

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## PS- 4

### Q.1

### Problem to demonstrate multicollinearity

**Consider the Credit data in the ISLR library. Choose balance as the response and Age, Limit and Rating as the predictors**

```
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.4.3

library(stargazer)

##
## Please cite as:
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
## Statistics Tables.
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

library(car)

## Warning: package 'car' was built under R version 4.4.3
## Loading required package: carData
## Warning: package 'carData' was built under R version 4.4.3
```

```

data("Credit")
str(Credit)

## 'data.frame':    400 obs. of  12 variables:
## $ ID      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ Income   : num  14.9 106 104.6 148.9 55.9 ...
## $ Limit    : int  3606 6645 7075 9504 4897 8047 3388 7114 3300 6819 ..
.
## $ Rating   : int  283 483 514 681 357 569 259 512 266 491 ...
## $ Cards    : int  2 3 4 3 2 4 2 2 5 3 ...
## $ Age      : int  34 82 71 36 68 77 37 87 66 41 ...
## $ Education: int  11 15 11 11 16 10 12 9 13 19 ...
## $ Gender   : Factor w/ 2 levels " Male","Female": 1 2 1 2 1 1 2 1 2 2
...
## $ Student  : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1 1 1 1 2 ...
## $ Married  : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 1 1 1 1 2 ...
## $ Ethnicity: Factor w/ 3 levels "African American",...: 3 2 2 2 3 3 1 2
3 1 ...
## $ Balance  : int  333 903 580 964 331 1151 203 872 279 1350 ...

head(Credit)

##   ID  Income Limit Rating Cards Age Education Gender Student Married Et
##   1   14.891  3606    283     2  34          11  Male      No      Yes Ca
##   2   106.025  6645    483     3  82          15 Female    Yes      Yes
##   3   104.593  7075    514     4  71          11  Male      No      No
##   4   148.924  9504    681     3  36          11 Female    No      No
##   5    55.882  4897    357     2  68          16  Male      No      Yes Ca
##   6    80.180  8047    569     4  77          10  Male      No      No Ca
##   Balance
##   1      333
##   2      903
##   3      580
##   4      964
##   5      331
##   6     1151

```

We have been specified to keep:

Response → Balance

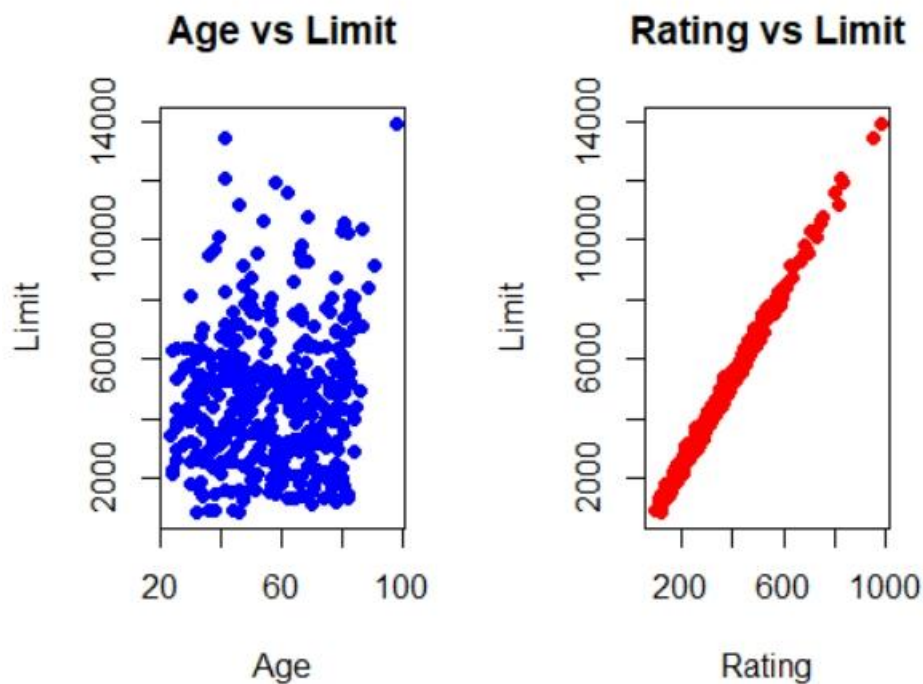
Predictors → Age, Limit, Rating

- (a) Make a scatter plot of (i) Age versus Limit and (ii) Rating Versus Limit. Comment on the scatter plot.

```
par(mfrow=c(1,2))

#(i) Age vs Limit
plot(Credit$Age, Credit$Limit,
     main="Age vs Limit",
     xlab="Age",
     ylab="Limit",
     col="blue", pch=19)

#(ii) Rating vs Limit
plot(Credit$Rating, Credit$Limit,
     main="Rating vs Limit",
     xlab="Rating",
     ylab="Limit",
     col="red", pch=19)
```



### Interpretation

- (i). The scatter appears random, there is proper pattern observed and there is weak collinearity between the two.
- (ii). The scatter appears linear, we see a strong positive relationship between Rating and Limit.

- (b) Run three separate regressions: (i) Balance on Age and Limit (ii) Balance on Age, Rating and Limit (iii) Balance on Rating and Limit. Present all the regression output in a single table using stargazer. What is the marked difference that you can observe from the output?





```
## -----
## Observations          400          400
400
## R2                    0.750          0.754
0.746
## Adjusted R2           0.749          0.752
0.745
## Residual Std. Error   230.532 (df = 397)    229.080 (df = 396)
232.320 (df = 397)
## F Statistic           594.988*** (df = 2; 397) 403.718*** (df = 3; 396) 5
82.820*** (df = 2; 397)
## =====
=====
## Note:                                                         *p<0
.1; **p<0.05; ***p<0.01
```

### Interpretation

In the first 2 models, Age is highly significant at 1% level implying Age has a statistically strong effect on balance. Age has negative impact on Balance since the coefficients of Age in both the models is negative.

In the first model, Limit was highly significant at 1% level but in second and third model where a new predictor Rating was introduced, Limit became statistically insignificant. This suggests, there exists multicollinearity between Rating and Limit. When both enter the model, Rating takes up most explanatory power and remains important even when Limit is included. This means Rating is a stronger determinant of balance than Limit.

(c) Calculate the variance inflation factor (VIF) and comment on multicollinearity.

The Variance Inflation Factor (VIF) reveals the presence and severity of multicollinearity across the models.

```
vif(model_1)

##      Age      Limit
## 1.010283 1.010283

vif(model_2)

##      Age      Rating      Limit
## 1.011385 160.668301 160.592880

vif(model_3)

##      Rating      Limit
## 160.4933 160.4933
```

## Interpretation

In the Age and Limit model, both Age and Limit have VIF values close to 1 ( $\approx 1.01$ ), indicating no multicollinearity meaning the predictors are almost completely independent of each other, and their coefficient estimates are stable and reliable.

However, once Rating is introduced in the Age, Limit and Rating model, the VIF values for Limit ( $\approx 160.59$ ) and Rating ( $\approx 160.67$ ) increase significantly, while Age remains near 1 ( $\approx 1.01$ ).

Similarly, in the Limit and Rating model, both variables again show extremely high VIF values ( $\approx 160.49$ ). A VIF exceeding 10 is typically considered serious; values above 100 indicate extreme multicollinearity. This means Limit and Rating are almost perfectly linearly correlated, causing inflated standard errors and making individual coefficient estimates unstable and statistically unreliable. This explains why Limit becomes insignificant when Rating is added in the model. In contrast, Age remains unaffected because it is not strongly correlated with the other predictors.

Overall, the output provides strong statistical evidence of severe multicollinearity between Limit and Rating, confirming the pattern observed in the scatter plot and regression table.

---

## Q.2

### Problem to demonstrate the detection of outlier, leverage and influential points

**Attach “Boston” data from MASS library in R. Select median value of owneroccupied homes, as the response and per capita crime rate, nitrogen oxides concentration, proportion of blacks and percentage of lower status of the population as predictors.**

```
library(MASS)
data("Boston")
str(Boston)

## 'data.frame':    506 obs. of  14 variables:
## $ crim   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ zn     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ indus  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ chas   : int   0 0 0 0 0 0 0 0 0 0 ...
## $ nox    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524
0.524 ...
## $ rm     : num  6.58 6.42 7.18 7 7.15 ...
## $ age    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ dis    : num  4.09 4.97 4.97 6.06 6.06 ...
## $ rad    : int   1 2 2 3 3 3 5 5 5 5 ...
```

```
## $ tax      : num  296 242 242 222 222 222 311 311 311 311 ...
## $ ptratio: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ black   : num  397 397 393 395 397 ...
## $ lstat   : num  4.98 9.14 4.03 2.94 5.33 ...
## $ medv    : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

`head(Boston)`

```
##      crim zn indus chas   nox    rm  age    dis rad tax ptratio  black
lstat
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900    1 296    15.3 396.90
4.98
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671    2 242    17.8 396.90
9.14
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671    2 242    17.8 392.83
4.03
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622    3 222    18.7 394.63
2.94
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622    3 222    18.7 396.90
5.33
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622    3 222    18.7 394.12
5.21
##      medv
## 1 24.0
## 2 21.6
## 3 34.7
## 4 33.4
## 5 36.2
## 6 28.7
```

We have been specified to keep:

Response → medv (median value of owner-occupied homes in \$1000s)

Predictors → crim (per capita crime rate by town), nox (nitrogen oxides concentration (parts per 10 million), black (the proportion of blacks by town), lstat (lower status of the population)

The objective is to fit a multiple linear regression model of the response on the predictors.

$$\text{medv} = \beta_0 + \beta_1 \text{crim} + \beta_2 \text{nox} + \beta_3 \text{black} + \beta_4 \text{lstat} + \epsilon$$

```
boston_model = lm(medv ~ crim + nox + black + lstat, data=Boston)
summary(boston_model)
```

```
##
## Call:
## lm(formula = medv ~ crim + nox + black + lstat, data = Boston)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.564  -4.004  -1.504   2.178  24.608
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 30.053584   2.170839  13.844  <2e-16 ***
## crim        -0.059424   0.037755  -1.574   0.116
## nox          3.415809   3.056602   1.118   0.264
## black        0.006785   0.003408   1.991   0.047 *
## lstat       -0.918431   0.050167 -18.307  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.183 on 501 degrees of freedom
## Multiple R-squared:  0.5517, Adjusted R-squared:  0.5481
## F-statistic: 154.1 on 4 and 501 DF,  p-value: < 2.2e-16
```

With reference to this problem, detect outliers, leverage points and influential points if any.